



Features of geometric sequences

Task A: Identifying geometric sequences

1) Sort these sequences into **geometric**, arithmetic and 'other'.

Geometric sequences

b) 1, 2, 4, 8, 16, ... $\times 2$

d) -1, -2, -4, -8, -16, ... $\times 2$

e) 1, 4, 16, 64, ... $\times 4$

k) 10, 20, 40, 80, 160, ... $\times 2$

l) 10, 100, 1000, 10 000, ... $\times 10$

Arithmetic sequences

a) 1, 2, 3, 4, 5, ... $+1$

f) 4, 16, 28, 40, 52, ... $+12$

i) 10, 40, 70, 100, 130, ... $+30$

Other sequences

c) 1, 2, 5, 10, 17, ...
 $+1 +3 +5 +7$

g) 4, 16, 30, 46, 64, ...
 $+12 +14 +16 +18$

h) 10, 15, 20, 30, 40, ...
 $+5 +5 +10 +10$

j) 10, 20, 60, 240, 1200, ...
 $\times 2 \times 3 \times 4 \times 5$

Task B: Generating geometric sequences

1) Generate the first five terms of these **geometric sequences**.

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| a) 1 st term 1, common ratio of 3 | 1, 3, 9, 27, 81, ... |
| b) 1 st term 10, common ratio of 3 | 10, 30, 90, 270, 810, ... |
| c) 1 st term 5, ratio 1 : 3 between terms | 5, 15, 45, 135, 405, ... |
| d) 1 st term 3, a ratio 1 : 5 between terms | 3, 15, 75, 375, 1875, ... |
| e) 1 st term 64, a ratio 2 : 5 between terms | 64, 160, 400, 1000, 2500, ... |
| f) 1 st term 81, common ratio of $\frac{1}{3}$ | 81, 27, 9, 3, 1, ... |
| g) 1 st term 1875, a ratio 5 : 1 between terms | 1875, 375, 75, 15, 3, ... |
| h) 1 st term 3, a ratio 3 : 1 between terms | 3, 1, $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{27}$, ... |

2) Alex is generating these **geometric sequences**.

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|---|-------------------------|
| a) 1 st term 216, common ratio $\times \frac{1}{3}$ | 216, 72, 24, 8, ... |
| b) 1 st term 216, common ratio $\times \frac{2}{3}$ | 216, 144, 96, 64, ... |
| c) 1 st term 216, common ratio $\times \frac{3}{2}$ | 216, 324, 486, 729, ... |



Because these multipliers are fractions, all of these will be decreasing sequences.

Show Alex that he is wrong.

Multiplier is greater than 1, therefore the sequence increases.

